Mohammed Afaaz

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github.com/mohammedafaaz afaazportfolio.vercel.app

Professional Summary

Aspiring AI/ML professional with experience in events organization and community leadership. Proficient in Python and Java, with a strong foundation in Data Structures and Algorithms, hands-on experience in backend development, and project work in machine learning and computer vision. Focused on building practical and scalable solutions.

Skills

Programming: Python, Java, C, C++, HTML, CSS, JS
Backend & APIs: RESTful APIs, HTTP, JSON, Authentication, Flask
Databases: SQLite, MySQL
Tools: Numpy, Pandas, Matplotlib, Git, GitHub, Streamlit, Power BI, LeetCode
Core Concepts: OOP, Data Structures and Algorithms, Machine Learning
Soft Skills: Leadership, Team Collaboration, Communication, Problem Solving

Training

Python Programming *Feb 2026*
Completed Python training at Ballari Institute of Technology and Management. Built a Flask-based web application to verify college bus pass validity using OpenCV, image processing, backend validation logic, and real-time authentication workflows.

Python Programming with Data Structures and Algorithms *Aug 2026*
Completed advanced Python training with a focus on Data Structures and Algorithms at Ballari Institute of Technology and Management, strengthening problem-solving skills through hands-on implementation of core data structures and algorithmic techniques.

Projects

EV Charging Slot Management System (Greedy Algorithm, Python) *Jun 2026*
Built a menu-driven Python application simulating an EV charging station's slot allocation using a Greedy Algorithm, featuring lowest-battery-priority allocation, a waiting queue for occupied slots, and duplicate booking prevention, with complexity analysis for core operations.

Malaria Cell Detection using CNN (Flask, Convolution Neural Network) *May 2026*
A full-stack web app to classify malaria-infected blood cells using a trained Convolutional Neural Network (CNN) and HTML, with Grad-CAM visual explanations for infected samples.

Student Registration Form (Java Swing GUI Application) *Jan 2026*
Developed a desktop-based Student Registration Form using Java Swing, implementing GUI components, ActionListener, and event handling for interactive form submission. Designed input validation and structured data handling to simulate a real-world registration workflow.

Hospital Bed Allocation System (DSA - Linked List in C) *Jan 2026*
Built a console-based hospital management system using Linked Lists in C to manage dynamic bed allocation and patient records. Implemented emergency and routine admissions, occupancy validation, and efficient deletion operations with dynamic memory management.

Pothole Detection and Reporting App (Flask, Machine Learning) *Dec 2025*
Developed an ML-powered web application using Python and YOLOv8 for real-time pothole detection and severity classification. Built a Flask REST API for image processing, automated severity assessment, and structured reporting with GPS-based location tracking.
2nd Runner-Up — SIH Internal Hackathon & IGNITRON CodeRush Hackathon 2025

VeriPass — Bus Pass Verification System (Flask, HTML5-QRCode) *May 2025*
Created a secure Flask web application to verify student bus passes using QR-based validation. Implemented backend authentication, real-time QR decoding, and database logging to ensure reliable and fraud-resistant pass management.

Leadership & Achievements

- Joint Secretary — AIML Department Forum (2025–2026)
- Webmaster — IEEE Student Branch, BITM (2025–2026)
- Secretary — IEEE CEDA/NT, BITM (2025)
- 2nd Runner-Up — SIH Internal Hackathon, BITM (2025)
- 2nd Runner-Up (Domain Wise) — IGNITRON, GMU Davangere (2025)
- Active Hackathon Participant

Certifications

- Infosys Springboard: Java Programming
- Udemy Code with Harry: Python Programming

Education

B.E. in Artificial Intelligence & Machine Learning, BITM *2024–2028*
CGPA: 8.6

Nandi PU College, Ballari *2022–2024*
Percentage: 94%

Nandi State School, Ballari *2012–2022*
Percentage: 92%